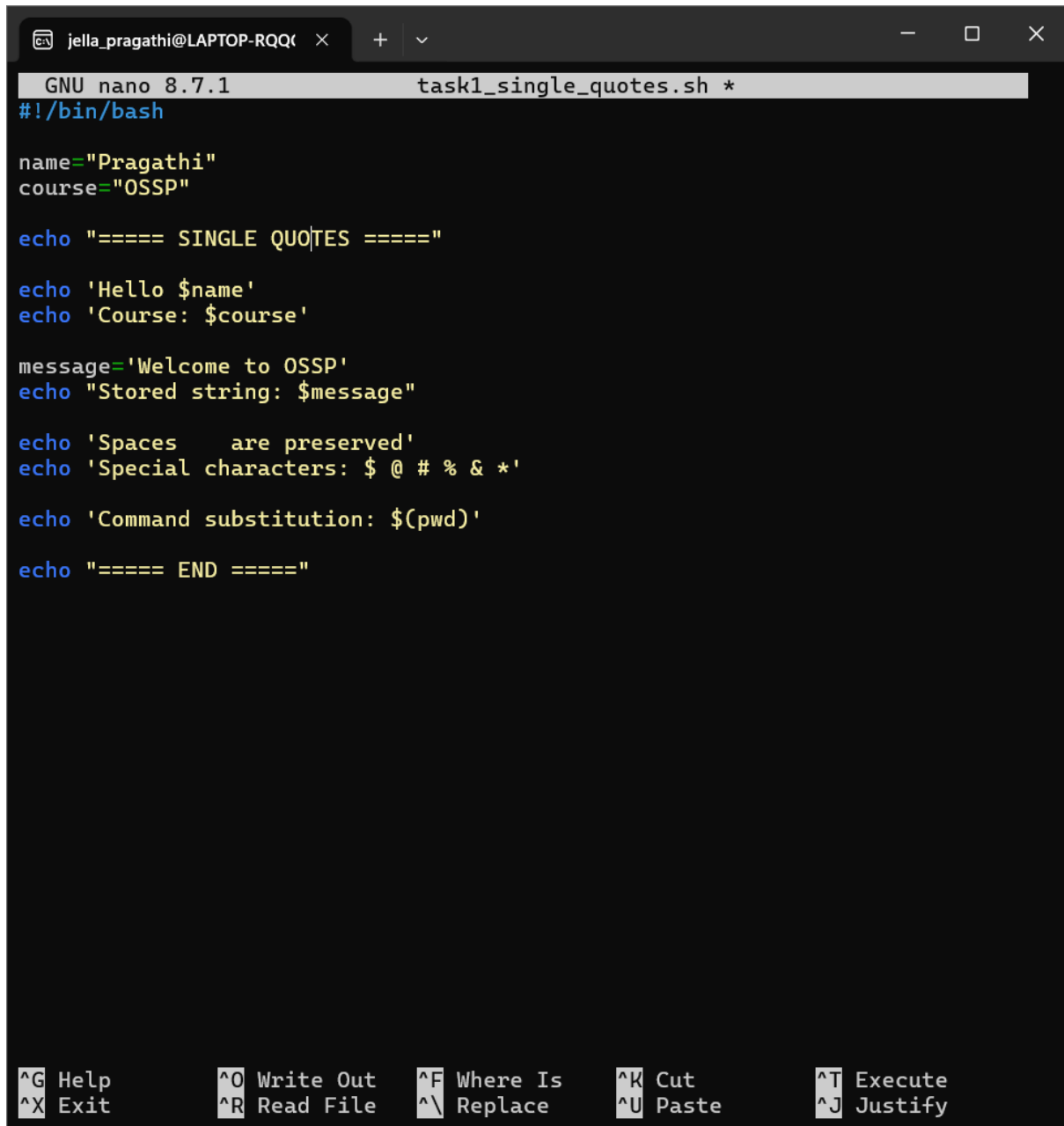


Skill Week 3

Session 5:

Task 1:

To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.



The screenshot shows a terminal window with a nano editor editing a file named `task1_single_quotes.sh`. The editor's status bar at the top indicates "GNU nano 8.7.1" and the file name. The script content is as follows:

```
#!/bin/bash

name="Pragathi"
course="OSSP"

echo "===== SINGLE QUOTES ====="

echo 'Hello $name'
echo 'Course: $course'

message='Welcome to OSSP'
echo "Stored string: $message"

echo 'Spaces    are preserved'
echo 'Special characters: $ @ # % & *'

echo 'Command substitution: $(pwd)'

echo "===== END ====="
```

At the bottom of the terminal, there is a row of keyboard shortcuts for the nano editor:

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify

```
jella_pragathi@LAPTOP-RQQGT555:~$ ^C
jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ nano task1_single_quotes.sh
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ chmod +x task1_single_quotes.sh
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./task1_single_quotes.sh
===== SINGLE QUOTES =====
Hello $name
Course: $course
Stored string: Welcome to OSSP
Spaces    are preserved
Special characters: $ @ # % & *
Command substitution: $(pwd)
===== END =====
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ |
```

Task 2:

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

```
jella_pragathi@LAPTOP-RQQG  ×  +  ▾
GNU nano 8.7.1 task2_double_quotes.sh *
#!/bin/bash

name="Pragathi"
course="OSSP"
age=19

echo "===== DOUBLE QUOTES ====="

echo "Hello $name"
echo "Course: $course"
echo "Age: $age"

message="Welcome to Shell Programming"
echo "Stored string: $message"

echo "This    contains    multiple    spaces"

echo "Current directory: $(pwd)"

command="ls"
echo "Command stored in variable: $command"

echo "Special characters: \$ @ # % & *"

echo "===== END ====="

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
=====
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ nano task2_double_quotes.sh
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ chmod +x task2_double_quotes.sh
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./task2_double_quotes.sh
===== DOUBLE QUOTES =====
Hello Pragathi
Course: OSSP
Age: 19
Stored string: Welcome to Shell Programming
This    contains    multiple    spaces
Current directory: /home/jella_pragathi/OSSP
Command stored in variable: ls
Special characters: $ @ # % & *
===== END =====
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ |
```

Session 6:

Task 1:

To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.



```
GNU nano 8.7.1 task1_escape.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[200];

    printf("Enter a command with escaped characters: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    printf("\nOriginal Input:\n");
    printf("%s\n", input);

    printf("\nParser Output:\n");

    for (int i = 0; input[i] != '\0'; i++) {
        if (input[i] == '\\' && input[i + 1] != '\0') {
            i++;
            printf("%c", input[i]);
        } else {
            printf("%c", input[i]);
        }
    }

    printf("\n");

    return 0;
}
```

Help Write Out Where Is Cut Execute
Exit Read File Replace Paste Justify

```

jella_pragathi@LAPTOP-RQQT555:~/OSSP$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ pwd
/home/jella_pragathi/OSSP
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ nano task1_escape.c
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ gcc task1_escape.c -o task1_escape
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./task1_escape
Enter a command with escaped characters: hello\ world

Original Input:
hello\ world

Parser Output:
hello world
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ^C
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./task1_escape
Enter a command with escaped characters: hello\@world

Original Input:
hello\@world

Parser Output:
hello@world
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./task1_escape
Enter a command with escaped characters: file\#name

Original Input:
file\#name

Parser Output:
file#name
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ |

```

Task 2:

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
jella_pragathi@LAPTOP-RQQQ x + v - □ ×
GNU nano 8.7.1 task2_child.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    printf("Parent Process\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        return 1;
    }

    if (pid == 0) {

        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Executing ls command...\n\n");

        execlp("ls", "ls", "-l", NULL);

        perror("Execution failed");
        exit(1);
    }

    else {

        printf("\nParent waiting for child...\n");

        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
        printf("\nParent waiting for child...\n");

        wait(NULL);

        printf("\nChild process completed.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
jella_pragathi@LAPTOP-RQQ...  +  v
file#name
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ nano task2_child.c
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ gcc task2_child.c -o task2_child
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./task2_child
Parent Process
Parent PID: 1118

Parent waiting for child...

Child Process
Child PID: 1119
Parent PID: 1118
Executing ls command...

total 136
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Aug 17 04:54 LICENSE
-rw-r--r-- 1 jella_pragathi jella_pragathi 265 Aug 17 05:03 Makefile
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Aug 17 04:54 README.md
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:02 assets
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 bin
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 docs
-rwxr-xr-x 1 jella_pragathi jella_pragathi 609 Aug 26 03:06 double_quotes.sh
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 include
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 obj
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:02 scripts
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16392 Aug 23 16:14 session4_wait
-rw-r--r-- 1 jella_pragathi jella_pragathi 1507 Aug 23 16:14 session4_wait.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 446 Aug 26 03:02 single_quotes.sh
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 src
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16224 Aug 26 03:26 task1_escape
-rw-r--r-- 1 jella_pragathi jella_pragathi 572 Aug 26 03:25 task1_escape.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 310 Aug 26 03:14 task1_single_quotes.sh
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16296 Aug 26 03:30 task2_child
-rw-r--r-- 1 jella_pragathi jella_pragathi 807 Aug 26 03:29 task2_child.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 419 Aug 26 03:16 task2_double_quotes.sh
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 tests
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16256 Aug 23 16:42 zombie
-rw-r--r-- 1 jella_pragathi jella_pragathi 780 Aug 23 16:41 zombie.c

Child process completed.
Parent process completed.
```